

Unit 3 questions

-> a company wants to deploy a 3-tier web application using Docker compose:

frontend-> react(node.js)

backend-> spring boot api

database-> postgresSQL

requirements:

all services must communicate:

environment variables should be used

services must start in dependency order

Answer:

```
project/
├── docker-compose.yml
├──
├── frontend/
│   ├── Dockerfile
│   └── (React app files)
├──
├── backend/
│   ├── Dockerfile
│   └── (Spring Boot jar)
```

-> create a docker compose setup for a node.js application connected to mongodb.

requirements:

node.js app runs on port 3000

mongodb runs on default port 27017

use environment variables for db connection

ensure mongodb starts before node.js

answer: node-mongo-app/

```
├── docker-compose.yml  ✓ (root folder)
├──
└── app/
    ├── Dockerfile      ✓ (inside app folder)
    ├── package.json
    └── server.js
```

dockerfile:

```
# Use Node base image
FROM node:18
# Set working directory
WORKDIR /app
# Copy package files
COPY package*.json ./
# Install dependencies
RUN npm install
# Copy all files
COPY . .
# Expose port
EXPOSE 3000
# Start app
CMD ["npm", "start"]
```

docker-compose.yml

```
version: '3.8'
services:
  mongodb:
    image: mongo:latest
    container_name: mongodb
    ports:
      - "27017:27017"
    volumes:
      - mongo_data:/data/db
  nodeapp:
    build: ./app
    container_name: nodeapp
    ports:
      - "3000:3000"
    environment:
      - MONGO_URI=mongodb://mongodb:27017/mydatabase
    depends_on:
      - mongodb
volumes:
  mongo_data:
```

package.json

```
{
  "name": "node-mongo-app",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "express": "^4.18.2",
    "mongoose": "^8.0.0"
  }
}
```

server.js

```
const express = require("express");
const mongoose = require("mongoose");
const app = express();
// Read environment variable
const mongoURI = process.env.MONGO_URI;
// Connect to MongoDB
mongoose.connect(mongoURI)
  .then(() => console.log("MongoDB connected"))
  .catch(err => console.log(err));
app.get("/", (req, res) => {
  res.send("Hello from Node.js with MongoDB!");
});
app.listen(3000, () => {
  console.log("Server running on port 3000");
});
```

```
/Users/madanavenkatachaitanyakrishna/Documents/int221/S0/vendor/ta-tikoma/phpunit-architecture-test/docker/docker-compose.yml
/Users/madanavenkatachaitanyakrishna/docker-wordpress/docker-compose.yml
madanavenkatachaitanyakrishna@chinnu-3 ~ % mkdir node-mongodb-app
madanavenkatachaitanyakrishna@chinnu-3 ~ % cd node-mongodb-app
madanavenkatachaitanyakrishna@chinnu-3 node-mongodb-app % touch docker-compose.yml
1
madanavenkatachaitanyakrishna@chinnu-3 node-mongodb-app % mkdir app
madanavenkatachaitanyakrishna@chinnu-3 node-mongodb-app % cd app
madanavenkatachaitanyakrishna@chinnu-3 app % touch Dockerfile package.json server.js
madanavenkatachaitanyakrishna@chinnu-3 app % nano docker-compose.yml
madanavenkatachaitanyakrishna@chinnu-3 app % nano package.json
madanavenkatachaitanyakrishna@chinnu-3 app % nano server.js
madanavenkatachaitanyakrishna@chinnu-3 app % code .
zsh: command not found: code
madanavenkatachaitanyakrishna@chinnu-3 app % docker-compose up --build
WARN[0000] /Users/madanavenkatachaitanyakrishna/node-mongodb-app/docker-compose.yml: the attribute `version` is obsolete, it
will be ignored, please remove it to avoid potential confusion
[+] Building 8.5s (12/12) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 588B 0.0s
```